

3.4 Geometry and measures

3.4.1 Properties and constructions

G1

Basic foundation content	Additional foundation content	Higher content only
use conventional terms and notations: points, lines, vertices, edges, planes, parallel lines, perpendicular lines, right angles, polygons, regular polygons and polygons with reflection and/or rotation symmetries use the standard conventions for labelling and referring to the sides and angles of triangles draw diagrams from written description		

G2

Basic foundation content	Additional foundation content	Higher content only
	use the standard ruler and compass constructions (perpendicular bisector of a line segment, constructing a perpendicular to a given line from/at a given point, bisecting a given angle) use these to construct given figures and solve loci problems know that the perpendicular distance from a point to a line is the shortest distance to the line	

Notes: including constructing an angle of 60° .

G3

Basic foundation content	Additional foundation content	Higher content only
apply the properties of angles at a point, angles at a point on a straight line, vertically opposite angles understand and use alternate and corresponding angles on parallel lines derive and use the sum of angles in a triangle (eg to deduce and use the angle sum in any polygon, and to derive properties of regular polygons)		

Notes: colloquial terms such as Z angles are not acceptable and should not be used.

G4

Basic foundation content	Additional foundation content	Higher content only
derive and apply the properties and definitions of: special types of quadrilaterals, including square, rectangle, parallelogram, trapezium, kite and rhombus and triangles and other plane figures using appropriate language		

Notes: including knowing names and properties of isosceles, equilateral, scalene, right-angled, acute-angled, obtuse-angled triangles. Including knowing names and using the polygons: pentagon, hexagon, octagon and decagon.

G5

Basic foundation content	Additional foundation content	Higher content only
	use the basic congruence criteria for triangles (SSS, SAS, ASA, RHS)	

G6

Basic foundation content	Additional foundation content	Higher content only
	apply angle facts, triangle congruence, similarity and properties of quadrilaterals to conjecture and derive results about angles and sides, including Pythagoras' theorem and the fact that the base angles of an isosceles triangle are equal, and use known results to obtain simple proofs	

G7

Basic foundation content	Additional foundation content	Higher content only
identify, describe and construct congruent and similar shapes, including on coordinate axes, by considering rotation, reflection, translation and enlargement	including fractional scale factors	including negative scale factors

G8

Basic foundation content	Additional foundation content	Higher content only
		describe the changes and invariance achieved by combinations of rotations, reflections and translations

Notes: including using column vector notation for translations. See also [G24](#)

G9

Basic foundation content	Additional foundation content	Higher content only
identify and apply circle definitions and properties, including: centre, radius, chord, diameter, circumference	including: tangent, arc, sector and segment	

G10

Basic foundation content	Additional foundation content	Higher content only
		apply and prove the standard circle theorems concerning angles, radii, tangents and chords, and use them to prove related results

Notes: including angle subtended by an arc at the centre is equal to twice the angle subtended at any point on the circumference, angle subtended at the circumference by a semicircle is 90° , angles in the same segment are equal, opposite angles in a cyclic quadrilateral sum to 180° , tangent at any point on a circle is perpendicular to the radius at that point, tangents from an external point are equal in length, the perpendicular from the centre to a chord bisects the chord, alternate segment theorem.

G11

Basic foundation content	Additional foundation content	Higher content only
solve geometrical problems on coordinate axes		

G12

Basic foundation content	Additional foundation content	Higher content only
identify properties of the faces, surfaces, edges and vertices of: cubes, cuboids, prisms, cylinders, pyramids, cones and spheres		

G13

Basic foundation content	Additional foundation content	Higher content only
interpret plans and elevations of 3D shapes	construct and interpret plans and elevations of 3D shapes	